UNIT 5 — 20 QUESTIONS

Motion Tracking & Object Recognition

Q81. The two fundamental steps of the Kalman filter are:

A. Detection and classification
B. Prediction and update (correction)
C. Sampling and quantization
D. Erosion and dilation

Q82. The Kalman filter is formally derived under the assumption that the system is:

A. Nonlinear with unknown noise
B. Linear with Gaussian noise
C. Discrete-colour only
D. Purely deterministic with no noise

Q83. The state vector in a typical Kalman-filter-based tracker often includes:

A. Only the object's colour
B. Position and velocity of the tracked object
C. The camera's focal length
D. The image's histogram

Q84. When a tracked object is briefly occluded, the Kalman filter:

A. Immediately loses all state information
B. Continues estimating position using prediction alone (no measurement update)
C. Switches to a different colour model
D. Recomputes the camera calibration

Q85. Observability, in the context of a tracking system, refers to:

A. The camera's field of view
B. The ability to infer the internal state from available measurements
C. The number of tracked objects
D. The frame rate of the video

Q86. The Kalman gain determines the balance between:

A. Two different cameras' outputs
B. Trust in the model's prediction versus trust in the new measurement
C. Colour and grayscale channels
D. Sampling and quantization

Q87. Compared to detecting an object independently in every frame, a filtering-based tracking approach is generally:

A. Less robust to noise and brief occlusion
B. More robust to noise and brief occlusion
C. Identical in performance
D. Only usable for static scenes

Q88. "Iterated estimation" in tracking refers to:

A. A single-pass, one-shot estimate
B. Repeated refinement of a state estimate over multiple iterations
C. Ignoring previous measurements entirely
D. Estimating only the camera's intrinsic parameters

Q89. Alignment-based object recognition primarily relies on:

A. Colour histogram matching only
B. Geometric correspondence between a model and the scene
C. Frequency-domain filtering
D. Kalman gain computation

Q90. A classic example of appearance-based object recognition is:
A. Canny edge detection
B. Eigenfaces (PCA-based face recognition)
C. Watershed segmentation
D. Zhang's calibration method

Q91. In the Eigenfaces method, Principal Component Analysis (PCA) is used to find directions that:
A. Minimize variance in the training data
B. Maximize variance in the training data
C. Minimize reprojection error only
D. Maximize colour saturation

Q92. Projecting a test image onto an eigenspace primarily achieves:
A. Increased image resolution
B. Dimensionality reduction while retaining key discriminative information
C. Removal of all noise permanently
D. Camera recalibration

Q93. Recognition invariants (e.g., to scale, rotation, illumination) are important because they help ensure:
A. Faster training only
B. Reliable recognition performance across varying real-world viewing conditions
C. Lower memory usage only
D. Simpler camera calibration

Q94. A well-known limitation of the classic Eigenfaces approach is its sensitivity to:
A. Image file format
B. Variations in pose and illumination
C. Camera frame rate
D. JPEG compression ratio only

Q95. In eigenspace-based recognition, classification of a test sample is typically performed using:
A. Watershed flooding
B. A nearest-neighbour distance metric in the reduced eigenspace
C. Canny edge detection
D. Optical flow estimation

Q96. In the linear state-space model used for Kalman filtering, the state transition equation includes a term for:
A. Process noise
B. Camera intrinsic parameters
C. Colour transform coefficients
D. SIFT descriptor length

Q97. The measurement equation in a Kalman filter relates:
A. Two different tracked objects to each other
B. The true state to the observed (noisy) measurement
C. The camera's rotation to its translation
D. Two consecutive eigenvectors

Q98. The Extended Kalman Filter (EKF) is used when:

A. The system is perfectly linear
B. The system dynamics or measurements are nonlinear
C. No noise is present in the system
D. Only appearance-based recognition is needed

Q99. An alignment-based recognition method verifies a match primarily by checking:
A. Colour histogram similarity only
B. Whether a geometric transformation consistently maps model features onto scene features
C. Frame rate consistency
D. Kalman gain convergence

Q100. A key general limitation of appearance-based (as opposed to invariant, model-based) recognition systems is:
A. They cannot run in real time under any condition
B. Sensitivity to conditions (pose, lighting, occlusion) not well represented in training data
C. They require no training data at all
D. They only work on grayscale images